

VANADIUM

(Data in metric tons, vanadium content, unless otherwise specified)

Domestic Production and Use: Vanadium production in Utah from the mining of uraniferous sandstones on the Colorado Plateau ceased in early 2020 and was not restarted in 2024. Secondary vanadium production continued in Arkansas, Delaware, Ohio, Pennsylvania, and Texas, where processed waste materials (petroleum residues, spent catalysts, and utility ash) were used to produce ferrovanadium, vanadium-bearing chemicals or specialty alloys, and vanadium pentoxide. Estimated U.S. apparent consumption of vanadium in 2024 decreased by 8% from that in 2023. Metallurgical use, primarily as an alloying agent for iron and steel, accounted for more than 90% of domestic reported vanadium consumption in 2024. Of the other uses for vanadium, the major nonmetallurgical use was in catalysts to produce maleic anhydride and sulfuric acid.

Salient Statistics—United States:	2020	2021	2022	2023	2024^e
Production from primary ore and concentrate	17	—	—	—	—
Production from ash, residues, and spent catalysts ^e	2,900	3,200	4,400	6,500	8,200
Imports for consumption:					
Aluminum-vanadium master alloy	101	35	104	221	110
Ash and residues ^{1, 2}	1,550	1,680	2,240	3,140	2,300
Ferrovanadium	1,360	2,170	2,650	2,280	1,800
Oxides and hydroxides, other	67	69	222	151	170
Vanadium chemicals ³	382	846	804	793	530
Vanadium metal ⁴	(⁵)	(⁵)	28	20	10
Vanadium ores and concentrates ¹	2	4	492	674	160
Vanadium pentoxide	1,670	1,710	1,980	2,320	2,500
Exports:					
Aluminum-vanadium master alloy	14	72	28	36	70
Ash and residues ¹	503	930	1,130	861	1,500
Ferrovanadium	210	173	154	159	70
Oxides and hydroxides, other	51	235	309	142	360
Vanadium metal ⁴	1	4	8	38	5
Vanadium ores and concentrates ¹	92	81	185	82	20
Vanadium pentoxide	50	17	143	28	120
Consumption:					
Apparent ⁶	7,110	8,200	11,000	14,800	14,000
Reported	7,920	8,030	7,510	^e 8,000	8,000
Price, average, vanadium pentoxide, ⁷ dollars per pound	6.47	8.17	9.29	7.50	5.45
Stocks, yearend ⁸	269	271	248	240	250
Net import reliance ⁹ as a percentage of apparent consumption	59	61	60	56	40

Recycling: Recycling of vanadium is mainly associated with reprocessing vanadium catalysts into new catalysts. The range in vanadium content in spent catalysts varies depending on the crude oil feedstock and the uncertainty associated with the quantity of vanadium recycled from spent chemical process catalysts was significant.

Import Sources (2020–23): Ferrovanadium: Canada, 48%; Austria, 37%; Russia, 7%; and other, 8%. Vanadium pentoxide: Brazil, 49%; South Africa, 35%; Russia, 7%; and other, 9%. Total: Canada, 34%; Brazil, 13%; Austria, 11%; South Africa, 11%; and other, 31%.

Tariff:	Item	Number	Normal Trade Relations 12–31–24
	Vanadium ores and concentrates	2615.90.6090	Free.
	Vanadium-bearing ash and residues	2620.40.0030	Free.
	Vanadium-bearing ash and residues, other	2620.99.1000	Free.
	Vanadium pentoxide, anhydride	2825.30.0010	5.5% ad valorem.
	Vanadium oxides and hydroxides, other	2825.30.0050	5.5% ad valorem.
	Ferrovanadium	7202.92.0000	4.2% ad valorem.
	Vanadium metal	8112.92.7000	2% ad valorem.
	Vanadium and articles thereof ¹⁰	8112.99.2000	2% ad valorem.
	Vanadium chemicals	(³)	5.5% ad valorem.

Depletion Allowance: 22% (domestic), 14% (foreign).

Government Stockpile: None.

Prepared by Désirée E. Polyak [(703) 648–4909, dpolyak@usgs.gov]

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Events, Trends, and Issues: The estimated average Chinese vanadium pentoxide (V_2O_5) price (98% V_2O_5 content) in 2024 was \$5.45 per pound compared with \$7.50 in 2023. The estimated United States ferrovanadium price (78%–82% vanadium content) was \$12.84 per pound in 2024 compared with \$16.42 in 2023. The World Steel Association estimated that global steel consumption increased by 1.7% in 2024. Total world steel production was estimated to have decreased by 1.5% during the first 7 months of 2024 compared with the same period in 2023. Like most ferroalloys, vanadium is largely dependent on the market characteristics of steel and specifically the Chinese steel industry. In 2024, China continued to be the world's top vanadium producer, producing most of its vanadium from vanadiferous iron ore processed for steel production.

Vanadium redox flow battery (VRFB) technology continued to be an increasingly important part of large-scale energy storage as it allows for high-safety, large-scale, environmentally friendly, medium- and long-term energy storage. Installations of VRFB projects continued to increase worldwide as energy companies looked to support renewable energy projects as many countries attempt to lower their carbon emissions. Many governments worldwide are promoting energy storage technologies, which creates favorable conditions for VRFB adoption. However, high capital and operating costs as well as limited vanadium feedstock availability remain the main drawback of VRFB technology. Despite the anticipated growth of VRFBs, there will be continued competition from a variety of alternative battery technologies looking to capture a portion of the energy storage market share.

World Mine Production and Reserves: Reserves for China and South Africa were revised based on company and Government reports.

	Mine production		Reserves ¹¹ (thousand metric tons)
	2023	2024 ^e	
United States	—	—	45
Australia	—	—	¹² 8,500
Brazil	5,420	5,000	120
China	^e 70,000	70,000	4,100
Russia	^e 20,000	21,000	5,000
South Africa	8,670	8,000	430
World total (rounded)	104,000	100,000	18,000

World Resources:¹¹ World resources of vanadium exceed 63 million tons. Vanadium occurs in deposits of phosphate rock, titaniferous magnetite, and uraniferous sandstone and siltstone, in which it constitutes less than 2% of the host rock. Significant quantities are also present in bauxite and carboniferous materials, such as coal, crude oil, oil shale, and tar sands. Because vanadium is typically recovered as a byproduct or coproduct, demonstrated world resources of the element are not fully indicative of available supplies.

Substitutes: Steels containing various combinations of other alloying elements can be substituted for steels containing vanadium. Certain metals, such as manganese, molybdenum, niobium (columbium), titanium, and tungsten, are to some degree interchangeable with vanadium as alloying elements in steel. Platinum and nickel can replace vanadium compounds as catalysts in some chemical processes. Currently, no acceptable substitute for vanadium is available for use in aerospace titanium alloys.

^eEstimated. — Zero.

¹Reported by the U.S. Census Bureau as kilograms of V_2O_5 . To convert V_2O_5 content to vanadium content, multiply by 0.56.

²Includes estimates for data suppressed by the U.S. Census Bureau in the years 2020 through 2024.

³Includes Harmonized Tariff Schedule of the United States codes for chloride oxides and hydroxides of vanadium (2827.49.1000), hydrides and nitrides of vanadium (2850.00.2000), vanadates (2841.90.1000), vanadium chlorides (2827.39.1000), and vanadium sulfates (2833.29.3000).

⁴Includes waste and scrap.

⁵Less than ½ unit.

⁶Defined as primary production + secondary production + imports – exports ± adjustments for industry stock changes.

⁷Chinese annual average V_2O_5 prices (98% V_2O_5 content). Source Argus Media Group, Argus Non-Ferrous Markets.

⁸Includes ferrovanadium, vanadium-aluminum alloy, other vanadium alloys, vanadium metal, vanadium pentoxide, and other specialty chemicals.

⁹Defined as imports – exports ± adjustments for industry stock changes.

¹⁰Aluminum-vanadium master alloy consisting of 35% aluminum and 64.5% vanadium and is the main master alloy for the vanadium industry.

Unwrought aluminum-vanadium master alloy (Harmonized Tariff Schedule of the United States code 7601.20.9030) was not included.

¹¹See Appendix C for resource and reserve definitions and information concerning data sources.

¹²For Australia, Joint Ore Reserves Committee-compliant or equivalent reserves were 3 million tons.